

(Bi<sub>3</sub>Pb), mp-559695 (Cs<sub>3</sub>BiF<sub>6</sub>), mp-561827 (Cs<sub>3</sub>TlF<sub>6</sub>), mp-1211553 (LuCrO<sub>5</sub>), mp-545399 (NaNiO<sub>6</sub>), mp-1522253 (NaSrSnBiO<sub>6</sub>), mp-1220399 (NdGaAu<sub>2</sub>), mp-1100068 (Sr<sub>2</sub>CoO<sub>4</sub>), mp-775149 (TiS<sub>2</sub>O<sub>8</sub>). We provide json files to visualize interactive UMAPs on [chemiscope.org](https://chemiscope.org).

### A.29.3 Analysis of highly-coordinated theoretical structures

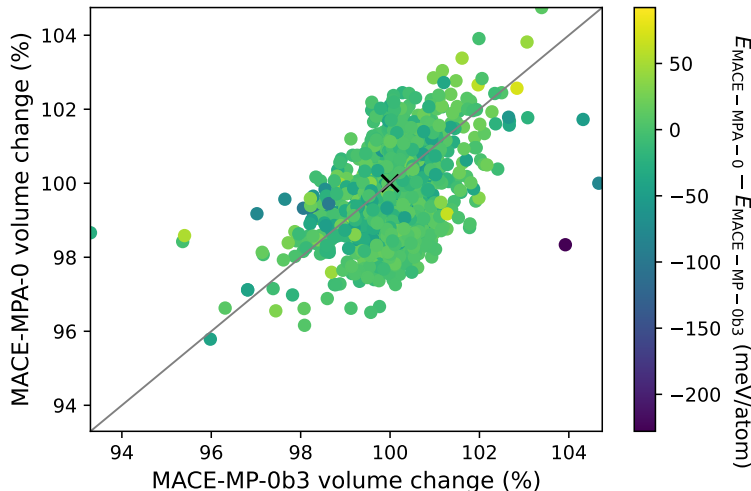


Figure S44: Comparison of the percentage volume change under relaxation relative to the initial structure volume as taken from the GNoME dataset. MACE-MP-0b3 is plotted on the horizontal axis, and MACE-MPA-0 on the vertical. The line indicates perfect agreement between both MACE models, and the “x” indicates perfect agreement with the GNoME volume. We observe a slight tendency of MACE-MP-0b3 to predict larger volume changes, and thus larger volumes, than MACE-MPA-0. The points are colored by the difference in energy between MACE-MPA-0 and MACE-MP-0b3, thus darker colors indicate MACE-MP-0b3 predicts a lower total energy than MACE-MPA-0, and brighter colors indicate that MACE-MP-0b3 predicts a higher total energy than MACE-MPA-0.

The recent GNoME database (99) published approximately 381,000 stable materials which could not be matched to any experimental structure in the Inorganic Crystal Structure Database (ICSD) (281), nor any public repository of theoretical structures such as the Open Quantum Materials Database (OQMD) or Materials Project. An analysis of the local coordination environment of these structures using the `CrystalNN` algorithm (282) reveals 21,300 structures with maximum coordination number greater than 16.

To evaluate the performance of MACE-MP-0b3 and MACE-MPA-0 against the GNoME potential which generated these structures, we have re-relaxed 1,199 structures with a predicted maximum coordination number greater than 20. Additionally, to compute formation energies, we re-relaxed all elemental structures in the Materials Project identified to be the lowest energy configuration from PBE/ +  $U$  calculations. The `FrechetCellFilter` class in the Atomic Simulation Environment (ASE) (275) was used to relax structures until the maximum (absolute) Cartesian component of any inter-atomic force was less than  $2 \times 10^{-2}$  eV/. This force convergence setting is consistent with the most stringent force relaxation criterion used by the Materials Project. No dispersion correction was used to augment the MACE models, consistent with the lack of a dispersion correction in GNoME. All relaxations ran successfully on CPU resources.

The MACE-relaxed structures are highly similar to the GNoME ones. S44 plots the percentage change in volume,

$$\delta V = (100\%) \times \frac{V_{\text{MACE}}}{V_{\text{GNoME}}} \quad (\text{S7})$$

for both MACE-MP-0b3 and MACE-MPA-0. MACE-MPA-0 tends to predict smaller volumes than MACE-MP-0b3, with a mean absolute deviation (MAD) from the GNoME structure volumes of  $0.117 \text{ \AA}^3/\text{atom}$ , compared to the  $0.085 \text{ \AA}^3/\text{atom}$  MAD of MACE-MP-0b3. However, their energies (indicated by the colors of the points) are nearly the same in all cases.

In virtually all cases, the MACE models relax structures to an indistinguishable configuration from their GNoME starting point. Only 14 (1) structures undergo an energy lowering of greater than 10 meV/atom during relaxation with MACE-MP-0b3 (MACE-MPA-0). 15 (14) structures underwent an increase of symmetry under relaxation with MACE-MP-0b3 (MACE-MPA-0), as measured by their space group number.

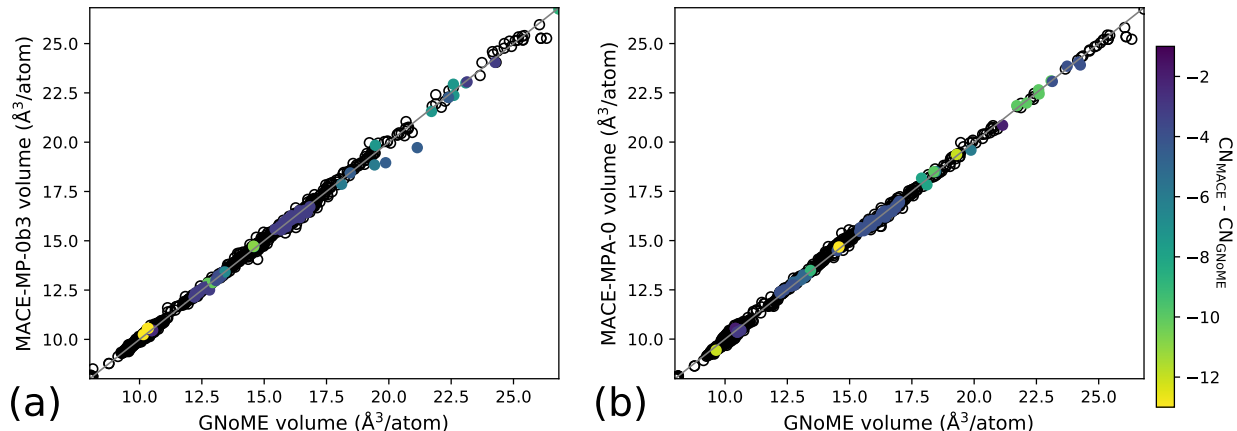


Figure S45: The final volume of the highly-coordinated GNoME materials before (horizontal) and after relaxation with a MACE model (vertical). Panel (a) plots the MACE-MP-0b3 model results, and (b) the MACE-MPA-0 results. The points are colored by their change in maximum observed coordination number as measured by *CrystalNN* (282): a hollow circle indicates no change. Nonzero changes in the maximum coordination number, suggesting a decrease in dense bond patterns, are indicated by increasingly light colors for larger magnitude changes. In all cases, the maximum coordination number was observed to decrease or stay the same.

Only one structure decreased in symmetry when relaxed with MACE-MP-0b3. The maximum coordination number decreased for 82 (87) structures under relaxation with MACE-MP-0b3 (MACE-MPA-0), and stayed the same otherwise. Figure S45 presents these observations for both MACE models.

Last, we consider the formation enthalpies, defined as

$$\delta H_f = \frac{1}{\sum_i c_i} \left[ H_{\text{solid}} - \sum_i c_i \mu_i \right], \quad (\text{S8})$$

the total enthalpy of a solid  $H_{\text{solid}}$  minus the chemical potentials of its elemental constituents  $\mu_i$  weighted by their stoichiometry  $c_i$  in the structure. Note that  $\sum_i c_i$  is the total number of atoms in the computational cell of the structure. After relaxation, the enthalpies and chemical potentials in Eq. S8 should be well-approximated by energies. Figure S46 presents the distribution of formation energies of the GNoME structures. Two distinct peaks can be observed, one for the intermetallics and covalently bound structures between -1 and 0 eV/atom, and one for the oxides below -1 eV/atom. Also note that no materials were found to be unstable as indicated by a positive formation energy.

Cumulatively, these observations suggest that both MACE models perform comparably to the GNoME UIP for these highly-coordinated structures, and yield equilibrated structures of equal or higher symmetry than the GNoME UIP.

### Similarity statement

By construction, the materials in the GNoME set do not exist in the MP dataset. The GNoME set was constructed in part by substituting elements on structures that originated in the Materials Project. However, some of the GNoME materials exist in the same chemical space as those in MP. For the 1,199 GNoME materials considered here, 131 (10.9%) contain exactly the same elements as structures in MP. The chemical space of maximum overlap, Cu-La-Zn, contains four GNoME (7ee54b7a37, 3422b9acb0, f36439bb45 and 68a5e7535f) and two MPtrj (mp-1223296 and mp-1093834) materials. We provide two JSON-format dictionaries, `MPtraj_chem_env.json.gz`, and `GNoME_chem_env_w_ovlp.json.gz`, which tabulate the chemical environments spanned by MP and the GNoME subset, respectively. The GNoME subset is further categorized by overlap with MP.

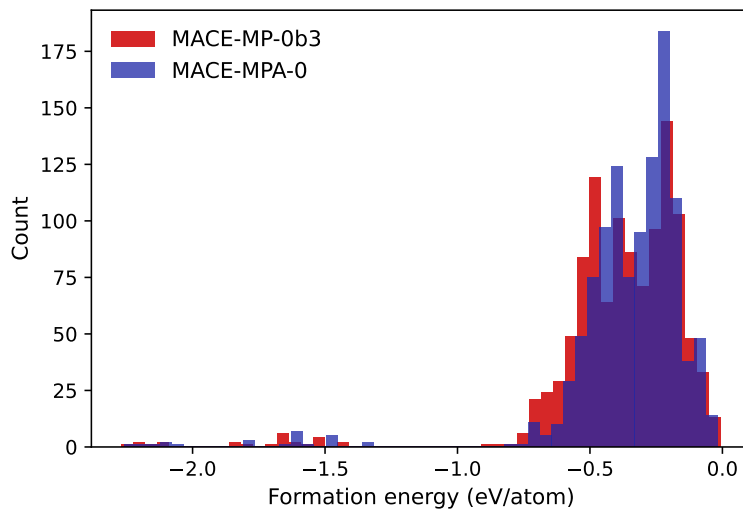


Figure S46: The distribution of formation energies (zero-pressure limit of the formation enthalpies at zero temperature) of the highly-coordinated GNoME structures after relaxation with MACE-MP-0b3(red) and MACE-MPA-0(transparent purple). Note the dual peaks in formation energies: between -1 and 0 eV/atom, most structures are intermetallics or covalently bound; below -1 eV/atom, most structures are oxides.

### Performance summary

Excellent energy prediction of WBM and GNoME hypothetical materials, and able to newly discover stable materials as validated by DFT.

### A.30 Alanine Tripeptide free energy surface

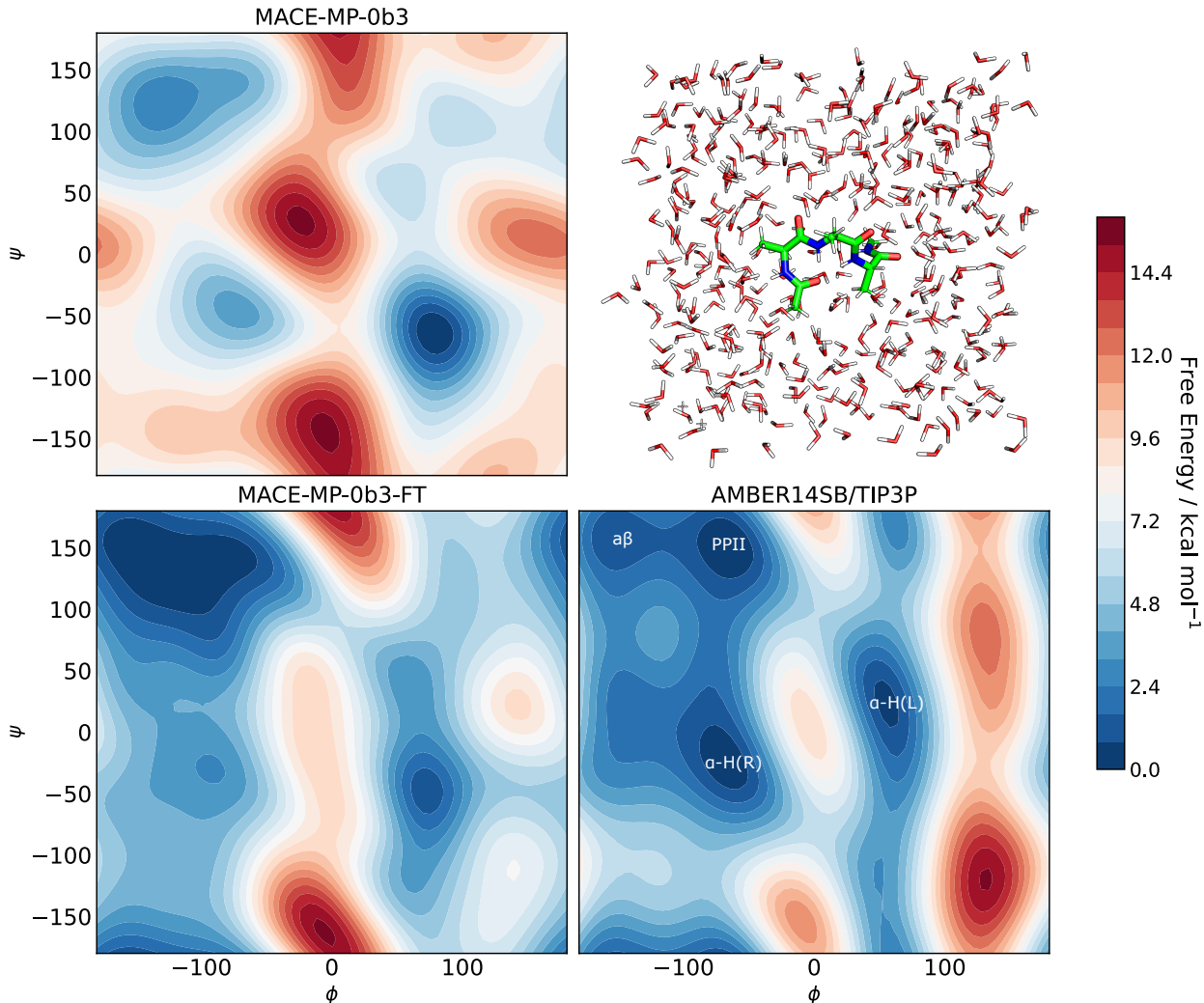


Figure S47: Free energy surfaces constructed from 1 ns sampling with well-tempered metadynamics. Enhanced sampling was performed on the central  $\psi$  and  $\phi$  backbone angles.

In this section, we investigate the ability of MACE-MP-0b3 to construct the free energy surface of a simple peptide. We simulate a periodic box containing an alanine tripeptide solvated with explicit water for 1 ns without D3 correction. Sampling of the central backbone torsions was accelerated by well-tempered metadynamics, as implemented in OpenMM, using an initial Gaussian height of 1 kJ/mol and a temperature scaling factor of 10. MACE-MP-0b3 only approximately identifies the features of the free energy landscape, whilst MACE-MP-0b3-FT identifies the positions of the minima in better qualitative agreement with the classical forcefield. MACE-MP-0b3-FT also more accurately predicts the relative population of the minima. However, neither model fully resolves the  $\alpha\beta$  and PPII minima around  $(\phi = -100^\circ, \psi = 150^\circ)$ . MACE-MP-0b3 in particular overpopulates the  $\alpha$ -H(L) basin at  $\phi = 80^\circ, \psi = -100^\circ$ .

#### Similarity statement

The MP dataset contains 99 structures containing exclusively the elements H, C, O and N. Based on UMAP analysis, we observe that all atomic environments in the example system are similar to environments in